

CIRCLE SCAN CONVERSION

$$(x - x_c)^2 + (y - y_c)^2 = r^2$$

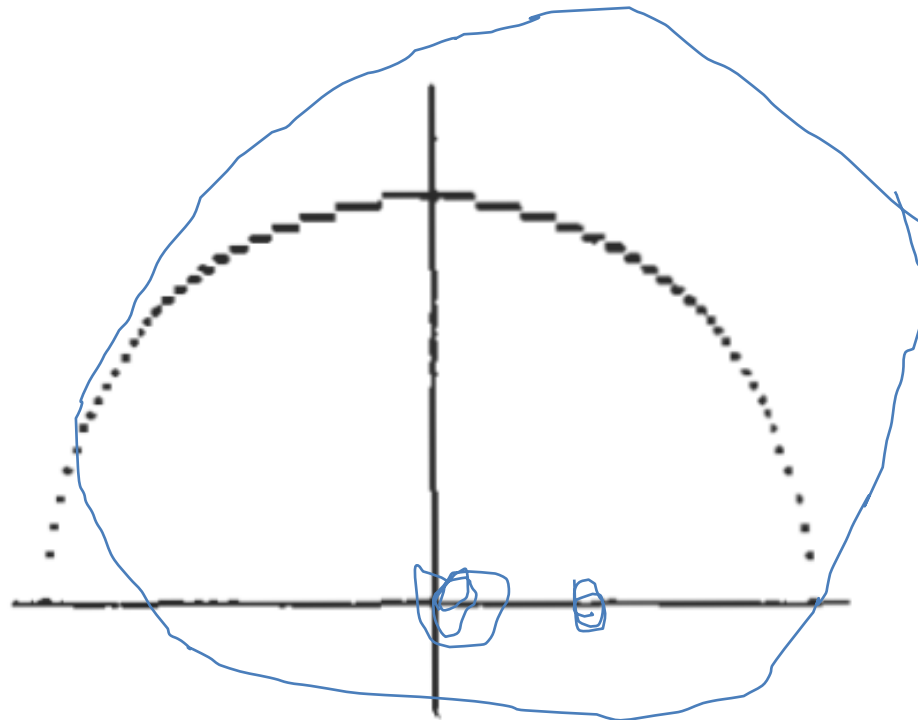
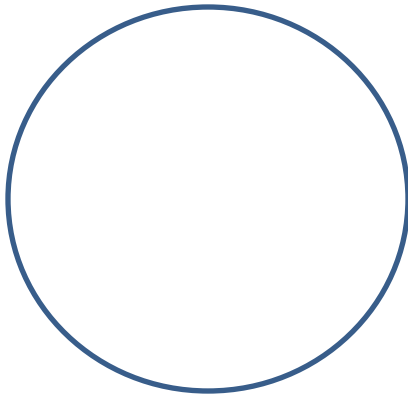
$$y = y_c \pm \sqrt{r^2 - (x_c - x)^2}$$

$$X_c = 0$$

$$Y_c = 0$$

$$X_0 = 0 \quad y_0 =$$

$$X_1 = 1 \quad y_1 =$$



$$x = x_c + r \cos \theta$$

$$y = y_c + r \sin \theta$$

Small angle = $0.1r$

$$X1 = r \cos (0)$$

$$Y1 = r \cos (0)$$

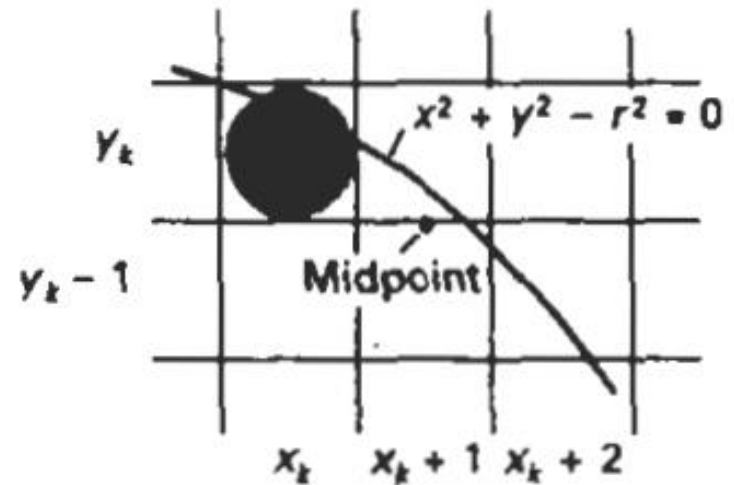
$$X2 = r \cos (0.1)$$

$$Y2 = r \cos (0.1)$$

Midpoint Circle Algorithm

$$f_{\text{circle}}(x, y) = x^2 + y^2 - r^2$$

$$f_{\text{circle}}(x, y) \begin{cases} < 0, & \text{if } (x, y) \text{ is inside the circle boundary} \\ = 0, & \text{if } (x, y) \text{ is on the circle boundary} \\ > 0, & \text{if } (x, y) \text{ is outside the circle boundary} \end{cases}$$



Any point (x, y) on the boundary of the circle with radius r satisfies the equation $f_{\text{circle}}(x, y) = 0$. If the point is in the interior of the circle, the circle function is negative. And if the point is outside the circle, the circle function is positive. To summarize, the relative position of any point (x, y) can be determined by checking the sign of the circle function:

$$\begin{aligned}
 p_k &= f_{\text{circle}}\left(x_k + 1, y_k - \frac{1}{2}\right) \\
 &= (x_k + 1)^2 + \left(y_k - \frac{1}{2}\right)^2 - r^2
 \end{aligned}$$

If $p_k < 0$, this midpoint is inside the circle and the pixel on scan line y_k is closer to the circle boundary. Otherwise, the midposition is outside or on the circle boundary, and we select the pixel on scanline $y_k + 1$.

$$\begin{aligned}
 p_{k+1} &= f_{\text{circle}}\left(x_{k+1} + 1, y_{k+1} - \frac{1}{2}\right) \\
 &= [(x_k + 1) + 1]^2 + \left(y_{k+1} - \frac{1}{2}\right)^2 - r^2
 \end{aligned}$$

the circle function at sampling position $x_{k+1} + 1 = x_k + 2$:

$$\begin{aligned} p_{k+1} &= f_{\text{circle}}\left(x_{k+1} + 1, y_{k+1} - \frac{1}{2}\right) \\ &= [(x_k + 1) + 1]^2 + \left(y_{k+1} - \frac{1}{2}\right)^2 - r^2 \end{aligned}$$

$$p_{k+1} = p_k + 2(x_k + 1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$$

where y_{k+1} is either y_k or y_{k-1} , depending on the sign of p_k .

Increments for obtaining p_{k+1} are either $2x_{k+1} + 1$ (if p_k is negative) or $2x_{k+1} + 1 - 2y_{k+1}$. Evaluation of the terms $2x_{k+1}$ and $2y_{k+1}$ can also be done incrementally as

$$2x_{k+1} = 2x_k + 2$$

$$2y_{k+1} = 2y_k - 2$$

At the start position $(0, r)$, these two terms have the values 0 and $2r$, respectively. Each successive value is obtained by adding 2 to the previous value of $2x$ and subtracting 2 from the previous value of $2y$.

The initial decision parameter is obtained by evaluating the circle function at the start position $(x_0, y_0) = (0, r)$:

$$\begin{aligned} p_0 &= f_{\text{circle}}\left(1, r - \frac{1}{2}\right) \\ &= 1 + \left(r - \frac{1}{2}\right)^2 - r^2 \\ p_0 &= \frac{5}{4} - r \end{aligned}$$

If the radius r is specified as an integer, we can simply round p_0 to

$$p_0 = 1 - r \quad (\text{for } r \text{ an integer})$$

since all increments are integers.

Midpoint Circle Algorithm

1. Input radius r and circle center (x_c, y_c) , and obtain the first point on the circumference of a circle centered on the origin as

$$(x_0, y_0) = (0, r)$$

2. Calculate the initial value of the decision parameter as

$$p_0 = \frac{5}{4} - r$$

3. At each x_k position, starting at $k = 0$, perform the following test: If $p_k < 0$, the next point along the circle centered on $(0, 0)$ is (x_{k+1}, y_k) and

$$p_{k+1} = p_k + 2x_{k+1} + 1$$

Otherwise, the next point along the circle is $(x_k + 1, y_k - 1)$ and

$$p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$$

where $2x_{k+1} = 2x_k + 2$ and $2y_{k+1} = 2y_k - 2$.

4. Determine symmetry points in the other seven octants.
5. Move each calculated pixel position (x, y) onto the circular path centered on (x_c, y_c) and plot the coordinate values:

$$x = x + x_c \quad y = y + y_c$$

6. Repeat steps 3 through 5 until $x \geq y$.

$p_0 = 1 - r$ radius is integer

Given a circle radius $r = 10$, we demonstrate the midpoint circle algorithm by determining positions along the circle octant in the first quadrant from $x = 0$ to $x = y$. The initial value of the decision parameter is

$$P_0 = 5/4 - 10 = -8.75 < 0$$

$$P_1 = p_0 + 2(1) + 1 = -8.75 + 2 + 1 = -5.75 < 0$$

k	p_k	(x_{k+1}, y_{k+1})	$2x_{k+1}$	$2y_{k+1}$
0	-9	(1, 10)	2	20
1	-6	(2, 10)	4	20
2	-1	(3, 10)	6	20
3	6	(4, 9)	8	18
4	-3	(5, 9)	10	18
5	8	(6, 8)	12	16
6	5	(7, 7)	14	14

$$P_2 = -5.75 + 2(2) + 1 = -0.75 < 0$$

$$P_3 = -0.75 + 2(3) + 1 = 6.25 > 0 \quad x_{k+1}, y_{k+1}$$

$$P_4 = 6.25 + 2(4) + 1 - 2(10) = -4.75 < 0$$

(0,10)

(1, 10)

(2, 10)

(3,10)

(4, 9)

(5, 9)

